

A Look Back at Project 1

Before we get started, let's take a look back at Project 1's `removeFirst` method.

- We'll compare a student solution with my solution:

A Look Back at Project 1

```
public T removeFirst() {  
    if(!isEmpty()){  
        T replace=sentinel.next.item;  
        sentinel.next=sentinel.next.next;  
        sentinel.next.prev=sentinel;  
        size--;  
        return replace;  
    }  
    return null;  
}
```

```
public T removeFirst() {  
    if (size == 0) {  
        return null;  
    }  
  
    Node oldFront = sentinel.next;  
    Node newFront = sentinel.next.next;  
  
    sentinel.next = newFront;  
    newFront.prev = sentinel;  
  
    size -= 1;  
    return oldFront.value;  
}
```

What do you like better about mine (on the right)?

Recommendations

Make your code obvious and easy to read.

- It's fine to create variables that serve only to give names to things (performance penalty is negligible or non-existent).

Use good style.

- Consistent spacing.
- Variable naming (camelCase in Java).
- Note: This will be mandatory starting a bit after the midterm.

```
public T removeFirst() {  
    if (size == 0) {  
        return null;  
    }  
  
    Node oldFront = sentinel.next;  
    Node newFront = sentinel.next.next;  
  
    sentinel.next = newFront;  
    newFront.prev = sentinel;  
  
    size -= 1;  
    return oldFront.value;  
}
```

Today's lecture is not in scope for midterm 1.

- But the ideas are useful, especially when we get to discussing TreeMaps, TreeSets, and Priority Queues later in the course.



Lecture 10

Subtype Polymorphism, Comparators, Comparables

CS61B, Spring 2026 @ UC Berkeley

Josh Hug and Kay Ousterhout

Polymorphism vs. Function Passing

Lecture 11, CS61B, Spring 2026

Polymorphism vs. Function Passing

Comparable and Comparators

- Comparables
- Compilation Error Puzzle
- Comparators
- Comparables vs. Comparators

Summary

Today we're going to dig deeper into inheritance.

- We'll see the contrast between **polymorphism** and **function passing**.
- Unlike Python, where function passing is common, most idiomatic Java code relies more heavily on polymorphism.

Let's see some examples in Python.

Polymorphism Example in Python

```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size
    def __gt__(self, other):
        return self.size > other.size
```

```
hadi = Dog("hadi", 30)
zora = Dog("zora", 44)

# uses the __gt__ function
print(hadi > zora)
```

In the code above, the **Dog** class overloads the > operator.

- Operator overloading is a form of **polymorphism**.
- Polymorphism: “the ability in programming to present the same programming interface for differing underlying forms” [[wiki](#)]
- Idea: We can use the > operator to compare anything (i.e. “any underlying form”) in Python, including Dogs.
 - To define the behavior of >, we implement the function `__gt__(self, other)`

Function Passing Example in Python

```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size
    def __gt__(self, other):
        return self.size > other.size
```

```
def name_len(d):
    return len(d.name)

max(doglist, key=name_len)
```

In the code above, the `max` function accepts a key function.

- This is an example of **function passing**.

Function Passing Example in Python

```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size
    def __gt__(self, other):
        return self.size > other.size
```

```
def name_len(d):
    return len(d.name)

max(doglist, key=name_len)

# or using a lambda, do it on one line:
max(doglist, key=lambda dog: dog.size)
```

In the code above, the `max` function accepts a key function.

- This is an example of **function passing**.

```
class Dog:
    ...
    def __gt__(self, other):
        return self.size > other.size

hadi > zora
```

```
max(doglist, key=name_len)
```

Today: Polymorphism in Java

- Above, we see **polymorphism** on the left and **function passing** on the right.
- Unlike Python, where both are common, most idiomatic Java code relies more heavily on polymorphism.
 - Example: Want to specify how two things should be compared? Where Python uses key functions, Java (typically) uses inheritance!
 - Note: We won't even discuss function passing in Java in our course, but it does exist.

Comparables

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Polymorphism vs. Function Passing

Comparable and Comparators

- **Comparables**
 - Compilation Error Puzzle
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Summary

Dogs

Consider the dogs below.

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));
```



"Grigometh", size: 200



"Pelusa", size: 5



"Clifford", size: 9000

Dogs

To find the maximum, we can use `Collections.max`. Let's try this.

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
Dog maxDog = Collections.max(dogs);
```



“Grigometh”, size: 200



“Pelusa”, size: 5



“Clifford”, size: 9000

Dogs

To find the maximum, we can use `Collections.max`. Let's try this.

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
Dog maxDog = Collections.max(dogs);
```

This results in the incomprehensible error message below.

- The issue is that `max` doesn't know how to compare two `Dog` objects.

Required type: Collection <? extends T>

Provided: List <Dog>

reason: no instance(s) of type variable(s) T exist so that Dog conforms to Comparable<? super T>

Unlike Python, where we'd define a `__gt__` method to overload the `>` operator, in Java, we have to specify that a `Dog` is something that can be compared.

Question: How do we specify that a class has a certain capability in Java?

Unlike Python, where we'd define a `__gt__` method to overload the `>` operator, in Java, we have to specify that a `Dog` is something that can be compared.

Question: How do we specify that a class has a certain capability in Java?

- We must implement some interface.

The Comparable Interface

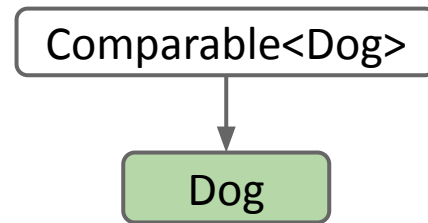
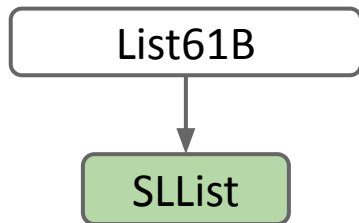
Unlike Python, where we'd define a `__gt__` method to overload the `>` operator, in Java, we have to specify that a `Dog` is something that can be compared.

In Java, we will declare to the world that a `Dog` is-a `Comparable`.

- Just like how earlier we said an `SLList` is-a `List61B`.

Since almost all of Java is written in Java, we can go look at `Comparable.java`.

- Link on github: [Comparable.java](#)



The Comparable Interface

Looking at [Comparable.java](#), we see:

```
public interface Comparable<T> {  
    /**  
     * Compares this object with the specified object for order.  
     * Returns a negative integer, zero, or a positive integer  
     * as this object is less than, equal to, or greater than the  
     * specified object.  
     * ...  
     */  
    public int compareTo(T o);  
}
```

Note: The full file is 142 lines long, almost all of which is documentation. This is not uncommon with really important parts of the library. Documentation is important.

Implementing Comparable

Let's implement the Comparable interface in Dog.java. This means:

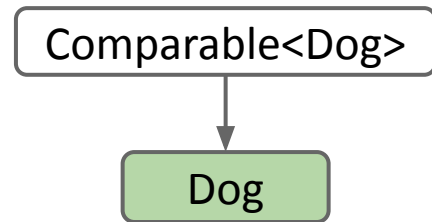
- Adding “implements Comparable<Dog>” to the top of the file.
- Overriding the compareTo method.

Implementing Comparable

Let's implement the Comparable interface in Dog.java. This means:

- Adding "implements Comparable<Dog>" to the top of the file.
- Overriding the compareTo method.

```
public class Dog implements Comparable<Dog> {  
    ...  
    @Override  
    public int compareTo(Dog uddaDog) {  
        if (size > uddaDog.size) {  
            return 1;  
        }  
        if (size < uddaDog.size) {  
            return -1;  
        }  
        return 0;  
    }  
}
```



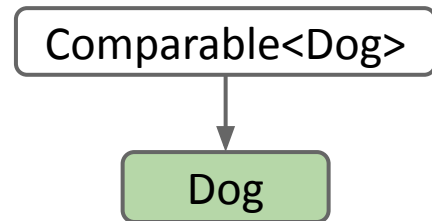
“uddaDog” courtesy of Harry Alu of [Ehu & Kai Adventures](#) (on the Big Island).

Implementing Comparable (Better Approach)

A cleaner implementation is shown below.

- This sort of code is very common in Java.

```
public class Dog implements Comparable<Dog> {  
    ...  
    @Override  
    public int compareTo(Dog uddaDog) {  
        return size - other.size;  
    }  
}
```



Now that we've implemented Comparable, the code below works fine!

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
Dog maxDog = Collections.max(dogs);
```

The flavor of polymorphism we've just employed is sometimes called **subtype polymorphism**.

- A supertype (Comparable) specifies the capability (in this case, comparison).
- A **subtype overrides** the supertype's abstract method.
 - Java decides what to do at runtime based on the type of the object that is invoking the method.

This is very different than Python's polymorphism, where there is a universal > operator which can be overloaded.

Compilation Error Puzzle

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Polymorphism vs. Function Passing

Comparable and Comparators

- Comparables
- **Compilation Error Puzzle**
- Comparators
- Comparables vs. Comparators

Summary

```
public class DogLauncher {  
    public static void main(String[] args) {  
        ...  
        Dog[] dogs = new Dog[]{d1, d2, d3};  
        System.out.println(Collections.max(dogs));  
    }  
}
```

```
public class Dog  
implements Comparable<Dog> {  
    ...  
    public int compareTo(Dog o) {  
        return this.size  
            - o.size;  
    } ...  
}
```

Q: If we omit `compareTo()`, which file will fail to **compile**?

- A. DogLauncher.java
- B. Dog.java
- C. Collections.java
- D. Comparable.java



```
public interface Collections {  
    // the max function  
    ...  
    int cmp = items[i].  
        compareTo(items[maxDex]);  
    ...  
} ...  
}
```

```
public class DogLauncher {  
    public static void main(String[] args) {  
        ...  
        Dog[] dogs = new Dog[]{d1, d2, d3};  
        System.out.println(Collections.max(dogs));  
    }  
}
```

```
public class Dog  
implements Comparable<Dog> {  
    ...  
    public int compareTo(Dog o) {  
        return this.size  
            - o.size;  
    } ...
```

Q: If we omit `implements Comparable<Dog>`, which file will fail to **compile**?

- A. DogLauncher.java
- B. Dog.java
- C. Collections.java
- D. Comparable.java



```
public interface Collections {  
    // the max function  
    ...  
    int cmp = items[i].  
        compareTo(items[maxDex]);  
    ...  
}...
```

Problem 1: Dog will fail to compile because it does not implement all abstract methods required by Comparable interface. (And I suppose DogLauncher will fail as well since Dog.class doesn't exist)

Problem 2: DogLauncher will fail, because it tries to pass things that are not Comparable, and Collections expects Comparables.

- If we used Override, Dog will not compile, because we have an override tag but we're not overriding (if we omit implements Comparable).

Comparators

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Polymorphism vs. Function Passing

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Summary

The term “Natural Order” is sometimes used to refer to the ordering implied by a Comparable’s compareTo method.

- Example: Dog objects (as we’ve defined them) have a natural order given by their size.



“Pelusa”, size: 5



“Grigometh”, size: 200



“Clifford”, size: 9000

Alternate Sorting

May wish to order objects in a different way.

- Example: By Name.



“Clifford”, size: 9000



“Grigometh”, size: 200



“Pelusa”, size: 5

Reminder: Key Functions in Python

In Python, we achieved the goal of specifying how to compare using **function passing**.

```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size
    def __gt__(self, other):
        return self.size > other.size
```

```
def name_len(d):
    return len(d.name)

max(doglist, key=name_len)
```

By contrast, in Java, idiomatic code uses **subtype polymorphism**.

The Comparator Interface

Java provides a Comparator interface for objects that are designed for comparing other objects.

```
public interface Comparator<T> {  
    int compare(T o1, T o2);  
    ...  
}
```

Comparator<Dog>

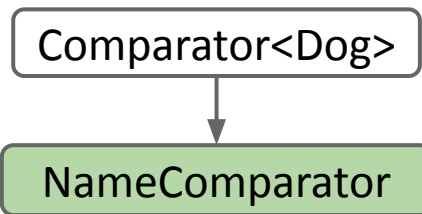
Let's try building and using one for comparing Dogs based on name.

Note: The Comparator interface has a LOT of default methods. We won't talk about them.

A Name Comparator

Java provides a `Comparator` interface for objects that are designed for comparing other objects.

```
public interface Comparator<T> {  
    int compare(T o1, T o2);  
    ...  
}
```



```
public class NameComparator implements Comparator<Dog> {  
    @Override  
    public int compare(Dog a, Dog b) {  
        return a.name.compareTo(b.name);  
    }  
}
```

What will be the output of the code below?

```
Dog a = new Dog("Frank", 1);  
Dog b = new Dog("Zeke", 1);  
Comparator<Dog> nc = new Dog.NameComparator();  
System.out.println(nc.compare(a, b));
```

```
public static class NameComparator implements Comparator<Dog> {  
    @Override  
    public int compare(Dog a, Dog b) {  
        return a.name.compareTo(b.name);  
    }  
}
```



- A. +1 B. Positive Number C. -1 D. Negative Number E. Zero

Comparator Quiz

What will be the output of the code below?

```
Dog a = new Dog("Frank", 1);  
Dog b = new Dog("Zeke", 1);  
Comparator<Dog> nc = new Dog.NameComparator();  
System.out.println(nc.compare(a, b));
```

```
public static class NameComparator implements Comparator<Dog> {  
    @Override  
    public int compare(Dog a, Dog b) {  
        return a.name.compareTo(b.name);  
    }  
}
```

The result will be a **negative number**: "Frank" is alphabetically less than "Zeke".

Using a Comparator to Find the Maximum

We can use our NameComparator to find the dog with the maximum name.

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
Dog maxNameDog = Collections.max(dogs, new Dog.NameComparator());
```

Compare with Python below (using length of name rather than name itself).

```
def length_of_name(dog):  
    return len(dog.name)  
dogs = [Dog("Grigometh", 10),  
        Dog("Pelusa", 5),  
        Dog("Clifford", 9000)]  
max(doglist, key=name_len)
```

This second argument
is an object of type
Comparator<Dog>

Using a Comparator to Find the Maximum

We can use our NameComparator to find the dog with the maximum name.

```
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```

Compare with Python below.

```
def length_of_name(dog):  
    return len(dog.name)  
dogs = [Dog("Grigometh", 10),  
        Dog("Pelusa", 5),  
        Dog("Clifford", 9000)]  
max(doglist, key=name_len)
```

In Java we package our comparison function inside of a Comparator object. We rely on **subtype polymorphism**.

In Python we often **pass functions** directly.

Awkward Instantiation

The NameComparator instantiation is awkward and aesthetically unpleasant (to me).

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
Dog maxNameDog = Collections.max(dogs, new Dog.NameComparator());
```

One fix is to add a static variable reference to a pre-instantiated NameComparator.

Awkward Instantiation

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```
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dogs.add(new Dog("Grigometh", 200));  
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dogs.add(new Dog("Clifford", 9000));  
Dog maxNameDog = Collections.max(dogs, new Dog.NameComparator());
```

One fix is to add a static variable reference to a pre-instantiated NameComparator. The usual naming convention is all caps:

```
public static NameComparator NAME_COMPARATOR = new NameComparator();
```

```
Dog maxNameDog = Collections.max(dogs, Dog.NAME_COMPARATOR);
```


A more modern approach is to use a lambda to define the Comparator.

- The lambda below defines and instantiates a Comparator object that compares the name of two dogs.
- We will not teach Java lambdas in 61B, nor will you be expected to learn them.

```
List<Dog> dogs = new ArrayList<>();  
dogs.add(new Dog("Grigometh", 200));  
dogs.add(new Dog("Pelusa", 5));  
dogs.add(new Dog("Clifford", 9000));  
  
Comparator<Dog> dc = (d1, d2) -> d1.name.compareTo(d2.name);  
Dog maxNameDog = Collections.max(dogs, dc);
```

Comparables vs. Comparators

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Polymorphism vs. Function Passing

Comparable and Comparators

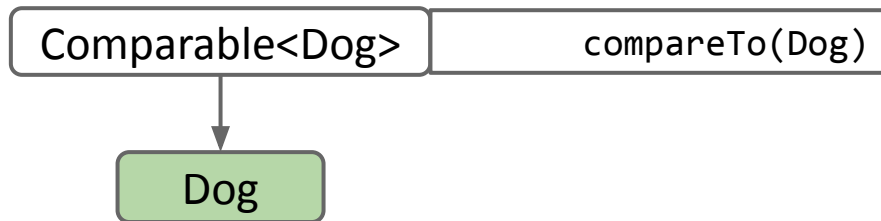
- Comparables
- Compilation Error Puzzle
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- **Comparables vs. Comparators**

Summary

Comparables vs. Comparators

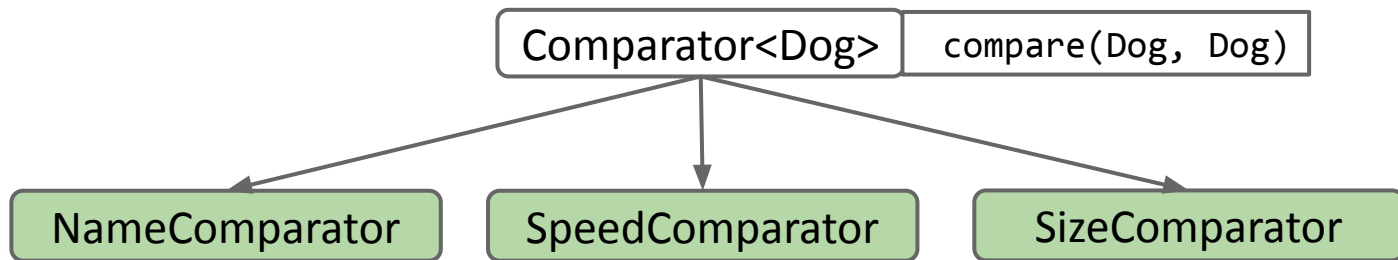
The Comparable interface specifies that a “natural order” exists.

- Instances of the class can compare themselves to other objects. Only one such order is possible.



The Comparator interface is used to compare extrinsically (by other classes).

- May have many such classes, each specifying one such order.



Comparables and Comparators do not have the same relationship as Iterables and Iterators.

- Comparable: I can be compared to another object.
 - `int compareTo(T other)`
- Comparator: I can tell you how to compare two objects.
 - `int compare(T x1, T x2)`
- Iterable: I can give you an iterator.
 - `Iterator<T> iterator()`
- Iterator: I can feed you objects.
 - `boolean hasNext()`
 - `T next()`

Summary

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Polymorphism vs. Function Passing
Comparable and Comparators

- Comparables
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Summary

Summary

Today we've seen **polymorphism** and **function passing**.

Comparable<T>

T

For comparing two objects using an intrinsic order:

- Python uses a form of polymorphism called **operator overloading**.
- Java instead uses **subtype polymorphism**.

Python is duck typed: Do not have to specify if > is available or not.

```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size

    def __gt__(self, other):
        return self.size > other.size

max(doglist)
```

```
public class Dog implements Comparable<Dog> {
    ...
    @Override
    public int compareTo(Dog uddaDog) {
        return size - other.size;
    }
}

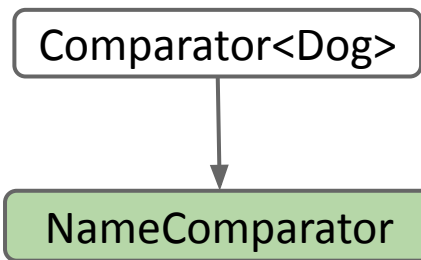
Collections.max(dogs);
```

Summary

Today we've seen **polymorphism** and **function passing**.

For comparing two objects using an alternate order:

- Python uses **function passing**. You provide a key function.
- Java instead (usually) uses **subtype polymorphism**, just like for intrinsic orders. We package the comparison function in a “Comparator” object.



```
class Dog:
    def __init__(self, name, size):
        self.name = name
        self.size = size

def name_len(dog):
    return len(dog.name)

max(doglist, key=name_len)
```

```
public static class NameComparator
    implements Comparator<Dog> {
    @Override
    public int compare(Dog a, Dog b) {
        return a.name.compareTo(b.name);
    }
}

Collections.max(dogs, new NameComparator());
```